

Toward a Variational Reconstruction of Janus-Type Geometric Theories

Author to be specified

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Abstract

We study the passage from geometric data to a variational field theory in the setting of Janus-type two-sided geometries. The central question is whether a geometry, a field content and a symmetry structure determine an action, or only constrain a class of admissible actions. We organize this question into independent tests involving finite-jet locality, invariant pairings, Helmholtz conditions, Noether identities, anomaly consistency and normalization. Parts of this architecture are represented by machine-checked Lean statements or by executable symbolic audits. The present work establishes a reproducible framework and several structural obstructions. It does not claim a unique renormalized Janus action or a derived absolute cosmological scale.

1 Introduction

A geometric picture is not yet a physical theory. A topology may suggest a field content, and a symmetry may suggest a Lagrangian, but neither suggestion is a derivation. The missing step is especially important in non-standard cosmological models, where geometric intuition can otherwise conceal free choices in the action, its couplings or its normalization.

This paper develops a disciplined reconstruction procedure. Rather than starting from a preferred Janus action, we ask which actions are compatible with explicitly stated geometric, local and variational assumptions. The procedure separates six logically different layers:

1. selection of an admissible action class;
2. local finite-jet dependence;
3. construction of invariant pairings;
4. Helmholtz and Noether compatibility;
5. anomaly consistency;
6. normalization and vacuum selection.

Passing one layer does not imply that the next one is satisfied.

Our contribution is therefore both positive and negative. We identify formal structures that can be checked, and we record obstructions that prevent an early uniqueness claim. This distinction is essential: a theorem about an abstract interface certifies an implication, but does not construct the physical object appearing in its hypotheses.

2 Variational reconstruction problem

Let M be a smooth manifold, let $\mathcal{F} \rightarrow M$ denote a field bundle and let $\mathcal{G}$ act on its sections. A local action has the schematic form

$$\mathcal{S}[\phi] = \int_M L(j^k \phi, g, \nabla, \dots) d\mu_g, \quad (1)$$

where $j^k \phi$ is the k -jet of the field and the omitted arguments denote the geometric structures admitted by the model. Its first variation is written

$$\delta \mathcal{S}[\phi](\psi) = \int_M \langle E(\phi), \psi \rangle d\mu_g + \int_{\partial M} \Theta(\phi, \psi), \quad (2)$$

with Euler source E and boundary term Θ .

The reconstruction problem runs in the opposite direction: given a candidate E , determine whether there exists a local L for which (2) holds, and determine how unique that L is. This requires more than covariance. The linearization DE_ϕ must satisfy the appropriate formal self-adjointness conditions, modulo gauge identities and boundary terms. Schematically, the Helmholtz condition reads

$$\langle DE_\phi(\psi_1), \psi_2 \rangle - \langle \psi_1, DE_\phi(\psi_2) \rangle = d_H B_\phi(\psi_1, \psi_2), \quad (3)$$

where d_H is the horizontal differential and B_ϕ is a boundary current. Equation (3) is a necessary local variationality test; sufficiency requires the stated cohomological and regularity hypotheses.

2.1 Interpretation

The distinction is physical. If a proposed field equation fails the Helmholtz test under the adopted assumptions, the problem is not merely that its Lagrangian has not yet been found: no such local Lagrangian exists in that class. If the test succeeds, the weaker conclusion is that a variational description is possible.

3 Action selection is an additional problem

Geometry and symmetry constrain the allowed scalar densities, but generally do not choose their coefficients. If $I_1, \dots, I_r$ are admissible local invariants, a family of actions may remain:

$$\mathcal{S}_\lambda[\phi] = \int_M \sum_{a=1}^r \lambda_a I_a(j^k \phi) d\mu_g. \quad (4)$$

The parameters λ_a may be constants, functions of invariant scalar data, or sections constrained by global descent.

Proposition 1 (Non-selection principle). *Geometric, symmetry and moduli data do not in general determine a unique action without a specified admissible field space, differential order, Hessian, critical configuration, normalization and boundary prescription.*

This proposition defines the burden of proof for uniqueness. A uniqueness statement is meaningful only relative to a declared action class.

4 Finite-jet locality and geometric reduction

The locality layer represents regular local natural operators through finite jets and holonomic prolongation. In the low-order development, the relevant source data reduce to geometric variables represented by a second fundamental form B and curvature-type data F , modulo the residual adapted-frame action. The corresponding local operator has the schematic form

$$\mathcal{O}_x = \Phi(B_x, F_x, \nabla B_x, \nabla F_x, \dots), \quad (5)$$

with Φ equivariant under changes of adapted frame.

Proposition 2 (Low-order reduction). *Under the regularity and naturality assumptions encoded in the formal development, the low-order local data admit a quotient represented by the relevant second-fundamental-form and curvature variables, subject to the residual frame action.*

The proposition is presently established only at its stated abstraction level. A physical Janus specialization still requires the global manifold and principal-bundle constructions, higher-order jet identification and effective descent.

4.1 Interpretation

If this locality program closes, a local observer needs only finite geometric data to formulate local laws. Global topology may still leave observable traces, but through identified holonomy, bundle or boundary effects rather than through unrestricted non-local terms.

5 Invariant pairings and global couplings

Let V_x and W_x be fibers carrying the isotropy action H_x . Pointwise admissible pairings lie in

$$\mathrm{Hom}_{H_x}(V_x \otimes W_x, \mathbb{R}). \quad (6)$$

Even when this vector space is one-dimensional at a regular point, the global coefficient multiplying a generator need not be a universal constant.

Proposition 3 (Global coupling qualification). *Pointwise multiplicity one does not imply a single constant global coupling. Global pairings form modules over invariant scalar functions, and their ranks may change across isotropy strata.*

This corrects a tempting but invalid inference from representation theory. Near a throat, defect or enhanced-symmetry locus, extension across strata must be proved rather than assumed.

5.1 Interpretation

A coupling that appears constant in one regime could depend on invariant geometric data or change character at exceptional strata. The proposition does not predict such variation; it identifies where constancy requires an additional argument.

6 Noether identities and anomaly filtering

For an infinitesimal gauge generator $R_\phi(\epsilon)$, invariance of the action implies a Noether identity of the form

$$R_\phi^\dagger E(\phi) = 0, \quad (7)$$

up to the declared boundary terms. Helmholtz compatibility and Equation (7) test classical consistency. Anomaly cancellation is independent: a classically consistent action can still fail after quantization. Program P therefore treats anomaly cancellation as a separate filter and not as a consequence of classical covariance.

7 Formal evidence and reproducibility

Claims are classified as follows: **T**, checked Lean theorem; **X**, executable symbolic or numerical audit; **C**, conditional theorem; **I**, analytic or geometric interface; **N**, no-go result or correction; and **O**, open construction. A conditional theorem is not promoted until its physical inputs have been constructed.

Table 1: Present evidence boundary.

Claim	Status	Current boundary
Geometry does not generally select an action	T/N	Relative uniqueness hypotheses are required
Low-order local reduction	T/X	Higher-order jet theorem remains open
Adapted-frame and SpinC structures	T/C/I	Global higher-dimensional packaging remains open
Global pairing coefficients form modules	T/C	Extension across strata needs hypotheses
Helmholtz/Noether reconstruction	C/I	Concrete Janus Euler source remains open
Unique renormalized action and absolute scale	O	Not claimed

The archival version will identify a frozen repository commit, the Lean and Mathlib versions, exact theorem-to-file correspondences and the commands used to reproduce each executable result.

8 Open completion conditions

A completed physical theory requires at least: (i) the actual Janus manifold and field bundles; (ii) higher-order locality and descent; (iii) a concrete compatible Euler source; (iv) nonlinear Helmholtz and Noether identities; (v) anomaly and boundary consistency; (vi) normalization and a stable vacuum; and (vii) predictions tested against independent observations.

These are dependencies, not rhetorical milestones. Failure at one stage is scientifically informative because it identifies which part of the proposed geometric picture cannot be promoted to a physical theory under the chosen assumptions.

9 Discussion

The present contribution is mathematical and methodological before it is phenomenological. It converts the passage from Janus geometry to dynamics into falsifiable obligations and prevents conditional formal statements from being mistaken for a completed cosmology.

If the remaining gates are eventually closed, the result would be more than an alternative description of the universe: it would connect two-sided geometry, local fields, conservation laws and quantum consistency in one constrained construction. At present this is an interpretation of the research direction, not an established physical conclusion.

10 Conclusion

We have formulated a variational reconstruction program for Janus-type geometric theories and isolated three structural conclusions: geometric data alone do not generally select a unique action; low-order local operators can be organized through controlled geometric quotients; and pointwise invariant pairings do not automatically yield constant global couplings. The framework also separates classical variationality, Noether consistency, anomalies and normalization. These distinctions provide a rigorous basis for future work on a concrete Janus action while keeping the current claim boundary explicit.